



MASTER 2 in COMPUTER SCIENCE
HUMAN-COMPUTER INTERACTION PROGRAM

Understanding Digital Music Production in the Times of Machine Learning

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Contents

Contents	i
Summary	ii
1 Introduction	1
2 Background	3
2.1 Algebraic Machine Learning (AML)	3
2.2 Music Creation	3
3 Related Work	5
3.1 Interactive Systems for Music Creation	5
3.2 AI in Music	7
4 Study	9
4.1 Methodology and Data Collection	9
4.2 Data Analysis	10
5 Study Evaluation and Implications	11
5.1 Results	11
5.2 Design Implications	12
6 Prototype Design and Implementation	15
6.1 Design Concept	15
6.2 Implementation	17
6.3 Scenario	26
7 Discussions	32
7.1 Choice between Gesture Interaction and Other Methods	32
7.2 Limitations	32
7.3 Future Work	32
8 Conclusion	34
9 Acknowledgments	35
Bibliography	36
A Appendix : Interview Questionnaire	40
B Appendix : Informed Consent Form	41
C Appendix : Additional Figures	44
D Appendix : Theoretical Overview of the Algebraic Machine Learning (AML) Model & Glossary	51

Summary

This thesis explores the understanding of digital music production tools, focusing on how increased transparency and intuitive design can improve the usability of digital music systems for musicians. As digital tools become increasingly integral to music creation, the ability for musicians to understand and interact effectively with these systems becomes critical. This research addresses the challenges associated with using complex software in creative workflows, in particular the need for tools that provide clear, understandable feedback and support for gesture-based interaction.

The research is based on a series of semi-structured interviews with seven experienced musicians to explore how current digital tools are used and the limitations they present in terms of user experience and workflow efficiency. Key findings from these interviews reveal musicians rely heavily on existing sounds and musical objects to explore rhythms and create music, they construct compositions channel by channel, they apply effects to individual channels, and they depend on visual feedback and system sensitivity for precise control.

These insights inform the design concept and development of a prototype system that incorporates gesture-based controls to provide a more natural and engaging way for musicians to interact with digital music production environments. The design implications for the development of future digital music tools are: 1. Allow hands-free gestures for creating and modifying musical elements. 2. Support both predefined and user-created gestures, enhancing adaptability. 3. Facilitate intuitive layering and structuring of sound elements. 4. Enable smooth operations across channels with enhanced customization options. 5. Improve visualization capabilities for real-time adjustments and precise control.

The findings highlight the importance of gestural interactions and the integration of explainability into the design of music production tools, not only to enhance user understanding, but also to support creative exploration and reduce the learning curve. The thesis concludes by proposing design principles for the development of future digital music tools that are more transparent, adaptable and responsive to the needs of contemporary musicians.

Keywords

Human-Computer Interaction, Music Creation, Gesture-Based Interaction, Algebraic Machine Learning, Explainable AI

Introduction

Music production has evolved significantly with the advent of digital tools, offering more control over aspects of a musician's compositions. However, the heart of music composition lies in the intuitive and spontaneous manipulation of sounds, rhythms, and effects. This work explores the role of digital tools in enhancing the creative process in music production, creation, and manipulation, with a particular focus on how the musicians interact with the tools to shape the soundscape.

The primary motivation of this research is to understand digital musical production and assist as designers to identify and bridge the gap between existing tools and the needs of users. Further, we also wish to explore the explainability of AI tools when understanding the results of different models and how having more explainability can improve user experience.

The research begins by asking a preliminary question: How can musicians utilize digital tools to create rhythms? This involves thematic analysis and interviews with musicians who have experience in digital music production. A music producer is defined as someone who can make music using digital tools, not just for manipulation, involving also creation. Furthermore, based on insights from these tools, the research explores design implications to provide the readers with guidance on designing better tools. It highlights the significance of rhythm manipulation, channel control, and iteration in composition.

Music production extends beyond merely arranging music or existing elements; it should involve a dynamic interaction between the composer and the tool. Hence, the research underlines physical interaction with music, including gestures and hand movements, has a potentially crucial place in shaping music. Whether the musician is isolating channels, stacking loops, or changing tempo and volume, the physical movements associated with all of these help a musician connect with their work, making the digital process more tangible and intuitive.

The work also speaks of the importance of visual and auditory feedback. The musician operates within the audio domain but has to use their eyes or visual cues as well to make decisions, whether in the mixing process or any kind of experimental phases within a composition. It provides musicians with the opportunity to know what works well together and what elements require further development.

The second research question explored is: How can a system be designed to create rhythms digitally? Through interviews with music production experts, we identify themes that inform the design implications for our system. By carefully considering the needs of contemporary musicians, we enhance their ability to compose new, expressive, and innovative compositions.

Next, the third research question is: How do we build the system? After conceptualizing the design, the design implications are implemented to develop the prototype system called Tune Crafter. This system allows users to manipulate music through gestures. This approach illustrates how musicians can utilize digital tools to create rhythms.

The development of the prototype system facilitates intuitive and expressive interactions in music production by enabling creative professionals to employ gestures, which are becoming increasingly prevalent in human-computer interaction (HCI), to explore music-based combinations. The system incorporates a range of activity recognition techniques to support natural and fluid interactions based on physical movements, including hand gestures. The MediaPipe

Hand Pose detection model, implemented through TensorFlow, facilitates the recognition of these gestures using a three-dimensional hand model. [32]

The system operates in two distinct scenarios: a fixed gesture system that utilizes a pre-existing library of MediaPipe's gesture dataset and a user-defined gesture system. In the first case, it's a comprehensive system with a guide to perform operations and has numerous use cases. In the latter case, users can train the algorithm with their own input gesture, thus enabling personalized and expressive interaction. The classification of these gestures is handled by Algebraic Machine Learning (AML) and other models, which enhances the system's interactivity. This framework accelerates the interaction cycles between the user and the AML model, thereby offering a more seamless and customizable creative experience for music production. The goal is to contribute to general ML literacy by allowing novices to train and explore Machine Learning approaches like AML, and to provide an easy tool for ML experts to explore and evaluate possibilities.

Background

2.1 Algebraic Machine Learning (AML)

The ALMA project investigates the potential of Algebraic Machine Learning (AML) in various applications, including music creation. The project aims to develop systems that can assist musicians by providing intuitive and explainable AI tools. Key components of the ALMA project include Marcelle[15], a system for music gesture interaction, and the integration of Human-Centered Design (HCD) methodologies.

Algebraic Machine Learning (AML)[17] is a new machine learning technique that is based on algebraic data representations. Unlike statistical learning, AML algorithms are parameter-free and therefore have the potential to be robust to statistical properties of the data. These systems are supposed to eliminate bias and discrimination in human-artificial intelligence systems, recall what they know when taught something new, and improve trust and reliability in human-artificial intelligence systems. As we take a look in the potential of explainability of such algebraic approaches in the context of music exploration. Professional musicians are a particularly demanding audience, who seek to generate new compositions or forms of musical expression but need a clear and predictable way to interact with new ML approaches. However, today's professional music tools focus primarily on implementing existing ideas, such as rendering notes onto a scale, rather than helping musicians to better understand how to generate new ideas with systems in a more fluid way. The ALMA project builds on the principles of instrumental interaction [12] and co-adaptation [16], as explored in earlier works, to create interactive systems that are discoverable, appropriable, and expressive, growing with the user to enhance rather than replace the user's skills. The project also leverages the concepts behind MayAI [14] and Marcelle [15] to advance the understanding and capabilities of AML in creative contexts such as music creation.

2.2 Music Creation

Understanding the basic tools and techniques used in digital music production is essential to understanding the context in which modern compositions are created. Digital audio workstations (DAWs) are central to music production, providing an all-in-one platform for recording, editing, mixing and mastering audio. Popular DAWs such as FL Studio[46], Ableton Live[47] and Logic Pro X offer a range of features that allow users to arrange musical elements, apply effects and automate processes. These platforms typically support both MIDI and audio recording. Audio processing tools are used to shape the sound within a composition. FabFilter Pro Q3[48], for example, is an equaliser that allows producers to fine-tune the frequency balance of their tracks. This tool is essential for adjusting tonal quality, managing the spectral content of different elements and ensuring clarity and balance in the final mix. Valhalla Vintage Verb[49] is another tool, providing reverb effects that add depth and space to the sound. The techniques of loops and sampling are fundamental to the processes of modern music production, particularly within the genres of electronic and hip-hop music[50]. A loop is a repeated section of sound that can form the basis of a rhythm or melody. Producers often manipulate these loops by changing

their pitch, time-stretching them, or applying effects to create variation and interest. Sampling involves the extraction of a portion of a sound recording and its reuse in a different track. This technique allows producers to integrate diverse sounds and influences into their compositions, often blending elements from different musical traditions or genres. MIDI controllers and virtual instruments are crucial components in digital music production[51]. MIDI controllers allow musicians to control virtual instruments and software synthesizers within a DAW, enabling real-time manipulation of sounds and adding expressiveness to productions, while virtual instruments, which are digital representations of acoustic and electronic instruments, provide a wide range of sounds and textures for music production[53]. Mixing and mastering are critical stages in music production[52]. Mixing involves balancing and adjusting individual tracks to create a cohesive and polished sound, including setting levels, panning, and applying effects. Mastering is the final step in music production, focusing on optimizing the overall sound for distribution across various playback systems. Equalization and Filtering give to the user the control over a track's frequency content through Equalization (EQ) and filtering[54]. By adjusting different frequency bands - bass, mid range and treble, producers can shape the tonal balance of a track. The inclusion of a filter further refines this control that allows drastic control of character of sound. Manipulations are essential for ensuring that each element occupies its intended space within the mix, thus achieving clarity and coherence in the final product. Advanced effects are the wide range of effects (also called FX) which are crucial for sound design. These effects include slicer, looper, crush, echo, flanger and reverb among others. For DJs, the ability to seamlessly mix tracks is important[55]. Features such as sync, hot cues and beats are also essential. Sync ensures that tracks play at the same tempo, allowing for smooth transitions. Hot Cues provide the ability to mark specific points in a track for instant recall during a performance. In addition, the ability to jump forward or backward by a specified number of beats (e.g. -8 or 8) allows DJs to navigate a track with agility.

Related Work

Previous research in interactive gesture interfaces, such as those presented at New Interfaces for Musical Expression (NIME), Designing Interactive Systems (DIS), and Conference on Human Factors in Computing Systems (CHI) conferences, has shown the potential of these systems to enhance musical expression. These interfaces allow musicians to interact with AI systems in more natural and intuitive ways.

3.1 Interactive Systems for Music Creation

Interactive Interfaces and Interfaces for Musicians

The computer for the 21st century, Weiser (1991); [27] : "The most profound technologies are those that disappear. They weave themselves into the fabric of everyday life until they are indistinguishable from it." The apple's new interface called Vision Pro is a good example of such a system. Spatial Computing is a phenomena where users interact with apps and data in their surrounding, creating a phenomena that feels immersive and intuitive. Here the device enhances our daily life without being the focus of our attention. Gestures allow humans to interact with headsets in a way that mimic the real world actions, such as grabbing, swiping or pointing. As no explicit interaction is necessary, it makes things less intrusive. This goes along with what Weiser remarks, "A good tool is an invisible tool. By invisible, I mean that the tool does not intrude on your consciousness; you focus on the task, not the tool."

When it comes to what kind of interaction an interface provides, we have moved from monitors to AR, and now we have more ways to interact, thanks to advances not only in technology but also in Human Computer Interaction and Design. The new wave of interfaces started in HCID with tangible interfaces and interactive systems.

Tangible User Interfaces and Interactive Systems, Ishii et al; (1997). [1] : Ishii and Ullmer's seminal work on Tangible Bits laid the foundation for tangible user interfaces (TUIs), emphasizing the seamless integration of digital and physical worlds. The paper's conclusion talks about the restrictions with current GUI systems in 1997, "Current GUI-based HCI displays all information as 'painted bit' on rectangular screens in the foreground, thus restricting itself to very limited communication channels. GUIs fall short of embracing the richness of human senses and skills people have developed through a lifetime of interaction with the physical world." This concept has profound implications in music creation tools where gestural interactions can control digital sound parameters and this way the digital space meets real space.

Mid Air Pan-and-Zoom on Wall-sized Displays, Nancel et al; (2011). [18] : The research work gives a foundational basis for researching different interaction techniques and why gesture interaction can be more efficient in some use cases. The experiment tested three key factors for the design of mid-air pan-and-zoom on desktop computers and studies of remote pointing. They identified three factors for the design of mid-air pan-and-zoom techniques: uni- vs. bi-manual interaction, linear vs circular movements, and level of guidance to accomplish the gestures in mid-air. There were a couple of surprising results. Gestures performed in free space tends to be

more prone to fatigue and generally less efficient than device-based input techniques. Adding guidance to input gestures increases, rather than decreases accuracy. Unimanual techniques perform honorably and depend more on context, one hand for a specific domain/task. Also, surprisingly higher efficiency of linear gestures when compared to circular, clutch-free gestures. This work aims to extend the research by applying these findings to gesture-based music production systems in the future, and to investigate how similar interaction principles can enhance the user experience in digital music environments. .

Knotty Gestures, Tsandilas et al; (2011). [19] : When it comes to thinking or improvising about the characteristics of what can be done with gestures, Knotty gestures provide users with a subtle, in-the-flow-of-writing technique for tagging information and subsequently interacting with the paper. Users can leave subtle marks that permit immediate interactions in the flow of writing and create rich opportunities for future interaction. The user must interact directly with the knot to discover and activate its functionality.

"When asked what musical instrument they play, there are not many computer music practitioners who would respond spontaneously with "I play the computer." This reflects the evolution of perspectives on digital instruments since the paper was first published in 2001. *Challenges in Musical Control of Computers, Wessel et al; (2001). [2]* : Wessel and Wright explored the intimate control musicians require over computer-based musical instruments. The research reflects on the importance of latency with gestures, "We place the acceptable upper bound on the computer's audible reaction to gesture at 10 milliseconds (ms) and the systems described in this paper provide for measured latencies nearer 7 ms." Further, They give a perspective to see music as a function of time, "Many musical gestures are continuous functions of time and should be treated as such, for example, the position along the string of a finger on a violinist's left hand." They identified key challenges in achieving expressive control, such as latency, resolution, and the complexity of mapping gestures to sound. Their insights are critical for developing music systems that offer nuanced, real-time control, enabling musicians to interact with digital instruments as naturally as with traditional ones.

How do we know if musicians are learning the system or finding it complicated? When is customization important, and when should the software adapt itself? Does the user adapt to the tool, or should the tool adapt to the user? These questions lead us to Wendy Mackay's thesis. *Users and Customizable Software: A Co-Adaptive Phenomenon, Mackay (2000); [16]* : It explores the intricate relationship between users and customizable software, framing it as a co-adaptive phenomenon. Mackay posits that the interaction between users and technology is dynamic, where both influence and adapt to each other over time. Co-adaptation allows users to modify software to suit their needs, which in turn influences future user behavior and software development. She examines how users not only adapt to technology but also appropriate it, leading to user innovations that can alter organizational practices. Key insights from Mackay's research include the identification of common customization patterns, such as the tendency for users to customize more when they first encounter new software and the influence of external events on customization behavior.

To gain insights into the kind of interface that would effectively support music creation, we sought a structured process to guide our exploration. The Reflexive Thematic Analysis framework, as outlined by Braun and Clarke, proved invaluable for understanding how individuals can enhance each other's work through collaborative improvement.[30] By identifying recurring themes, we were able to focus on key areas requiring attention.

DesignPrompt: Using Multimodal Interaction for Design Exploration with Generative AI,

Xiaohan et al; (2023). [31] : Building on this foundation, we further explored design implications derived from the identified themes, applying methods described in the paper Design Prompt. The research work by Xiaohan Peng, Janin Koch, and Wendy E. Mackay introduces DesignPrompt as a digital moodboard tool through which designers can embed images, colors, text, and semantics to form a single Generative AI (GenAI) prompt. A significant takeaway from this paper was the integration of design techniques that incorporate movement, much like gesture-based interaction. The use of multimodal input not only fosters exploration but also supports self-expression, serving as a metaphor for our design approach. A Usage study with 12 professional designers found that multimodal input supported exploration and self-expression in ways that were often unexpected, such as detailed multimodal prompts or novel design patterns that maintain consistency. An important contribution of this work to the thesis is its alignment with the same advisors and lab, Ex-Situ, and the continuation of the research trajectory established by Xiaohan. The interfaces in the thesis also support diverse interaction modalities, such as gestures, voice, etc. allowing for intuitive and flexible creative expression.

Collaborative Musical Experiences, Blaine et al; (2003). [4] : Blaine and Fels examined the social and collaborative aspects of musical performance, emphasizing the need for systems that support group interaction and real-time feedback. Their research states the importance of designing music systems that facilitate collaboration, enabling multiple users to interact simultaneously and creatively.

3.2 AI in Music

The MediaPipe Hand Pose detection model is available via Tensorflow[26] in this work to recognize hand gesture input based on a 3D hand-model.[32] A straightforward algorithm is utilized to calculate the gestures. The dataset comprises three distinct datasets. The datasets comprise an in-the-wild dataset, an in-house collected gesture dataset, and a synthetic dataset. The model comprises three outputs: firstly, 21 hand landmarks, comprising x, y, and relative depth coordinates. Subsequently, a hand flag is generated, indicating the probability of hand presence in the input image. Additionally, a binary classification of handedness is performed, e.g. left or right hand. The algorithmic process commences with the following steps. Initially, the state of each finger, e.g. bent or straight, is determined via the accumulated angles of joints. Subsequently, the set of finger states is mapped to a set of predefined gestures. This enables the estimation of basic static gestures with reasonable quality.

Machine Learning in Interaction: Tool, Material, Culture, Caramiaux (2023); [7] : Interactive Machine Learning (IML) has emerged as a significant approach that contrasts with traditional machine learning workflows by enabling rapid, focused, and incremental model updates in response to user input. This approach allows users to interactively examine the impact of their actions, facilitating a continuous dialogue between the user and the machine learning system. Such interactions are essential for domains like creative practices, where the subjective assessment of machine learning models by artists and designers is critical. Further, Baptiste's doctoral research explores the intersection of gestural interfaces and musical expression. His work explores the potential of three-dimensional hand movements as a means of intuitive musical control. Specifically, he explores how variations in gesture velocity and trajectory in 3D space can modulate musical parameters, fostering a dynamic and embodied approach to human-computer music interaction. The concept of machine learning as a design material

emphasizes the importance of exposing the internal workings of machine learning systems to users. By doing so, users can manipulate, assemble, and modify machine learning elements, transforming them into materials for creative exploration. The application of machine learning in creative practices, such as music and art, demonstrates the potential for these technologies to enable new forms of expression and innovation. For instance, gesture recognition systems used in artistic performances allow for the creation of motion-sound interactions, where machine learning models map gestural inputs to auditory outputs based on provided examples. This inductive approach to programming by demonstration has been instrumental in making machine learning accessible to non-experts in the creative industry.

Investigating Explainability of Generative AI for Code through Scenario-based Design, Sun et al (2022); [6] : Sun et al conducted 9 workshops with 43 software engineers in which real examples from state-of-the-art generative AI models were used to elicit user's explainability needs. Scenario-based design and question-driven design approach led to the identification of 11 categories of explainability needs in the context of GenAI for code. The procedure was quite informative, and so was the approach to get the design. Their work, methodology, and results inspires the thesis to enable better human-AI collaboration.

It is important to understand how users engage with these systems and the feedback they receive in the form of recommendations, to enhance the overall experience. *Explainability in Music Recommender Systems*, Fiebrink et al; (2016). [34] : The research work by Afchar et al. explored the importance of explainability in music recommender systems, emphasizing that transparent algorithms can improve user trust and satisfaction. Their research suggests that music recommendation systems should not only focus on providing accurate song suggestions but also on explaining why those particular songs are recommended. This approach can boost user engagement and trust in the system.

xAI for Musicians and Arts, Bryan-Kinns et al, (2023). [33] : The paper focuses on providing interface elements familiar to musicians and focuses on the real-time feedback relationship between musician and instrument. In this work, Algebraic machine learning is the xAI component where users can understand better the way the system understands. .

Study

4.1 Methodology and Data Collection

This section should provide detailed information about the research methods used in the study, including participant recruitment, study design, data collection procedures, and any other relevant details.

An in-depth semi-structured interview was conducted where 7 people were interviewed with backgrounds in music production, using mixing instruments and have performed live. The Participants were recruited using electronic means (Social Media like Instagram, Facebook) and word of mouth. For participants to be eligible for the study, they had to be (1) a Background in Playing Music (2) Knowledge of Mixing/Producing music using either tactile turn tables or digital mediums. In addition, the demographics data like age, gender, choice of musical instruments, years of experience with music of participants were collected.

The qualitative study was further analyzed using Thematic analysis and reflected on the findings to provide recommendations for the design of music technology that can better help people structure music production and playback.

Participant ID	Age	Gender	Choice of Musical Instruments	Choice of Digital Tools	Years of Experience
P1	23	Male	Piano	FL Studio	3
P2	24	Male	Guitar	Rekordbox & Pioneer DDJ 1000	5
P3	29	Male	Guitar	FL Studio	6
P4	29	Male	Guitar	FL Studio &	10
P5	24	Male	Guitar	FL Studio & Rekordbox	5
P6	22	Male	Drums	Abelton & FL Studio & Rekordbox	7
P7	25	Male	Guitar	FL Studio & Audacity	10

Table 4.1: Demographic and Tool Usage Data of Participants

I conducted in-depth individual interviews, each lasting between 45 minutes to 1 hour 15 minutes.

I began by a question with the participants what their workflow was for creating rhythms and how they do it. Next, they were to detail methods by which they found new sounds and rhythmic patterns; these include, for example, how they find new sounds or what makes them choose which sounds to implement in their projects. Then I also wanted to know about the kinds of tools and procedures they apply to change the existing rhythms that they might be having in

their work. Afterward, I asked the participants whether they ever used any system, for example, drum pads on a virtual environment or touch-sensitive surface, which helps to emulate physical interaction. I asked the participants if they could identify and explain what was easier and what was harder to do on these digital tools compared to the physical instruments. I then asked what kinds of feedback were most useful for participants, in general, from the visual, auditory, and haptic feedback sources, and how they used that to implement it into their process of making rhythms. I also examined how rhythm visualizations, e.g. waveform display or beat grids, would affect one's workflow and asked whether there were any particular types of visualization that were useful. I also inquired about how physical actions and physical experiences brought them through the creativity process and into music expression, including how different ways of moving were most natural or useful, less likely to be employed personally, and if anything was very hard and annoying. I then asked if there were tasks they nearly always did when generating rhythms and, if not, if there were things they did rarely. These were followed by related questions about why they did these things rarely. Finally, I investigated how the participants modified their workflows and tools in creating music, focusing on the aspects that they most often changed and the motivations behind such customizations.

4.2 Data Analysis

Informed consent was obtained via a consent form and permission to audio-record was also taken from each participant before the interview. All audio recordings were transcribed for analysis with GDPR compliant whisper tool.

It used a combination of open and axial coding, where each line was read and open coded each interview transcript line by line and then clustered codes based on themes in the data. We group similar codes to form high level themes, Done iteratively. This approach helps us structure our findings and the refinement of research questions. I made sure to maintain the privacy of participants while reporting quotes by anonymizing any personal data.

Study Evaluation and Implications

5.1 Results

The following section presents the findings of the results from the interview study. The findings are split into five themes.

Musicians use existing sounds and musical objects to explore rhythms and create music: Across all participants(7/7), a common characteristic is how instinctively they explore existing sounds and rhythms to create new musical pieces. Most participants emphasized that using pre-existing loops, samples, and beats creates the backbone of the composition. For example, P5 mentioned starting with a sample or loop that already exists and then building around it to create something new. Similarly, P1 and P3 noted that they use their sound libraries to find sounds and make loops as a starting point for making new tracks. This indicates that musicians often turn to what is known in order to build off of it creatively in a unique way. P1 mentioned building tracks layer by layer, starting with a base and adding various instruments. P2 elaborated on the ability to record multiple loops from different sources and switch them on and off as needed. P3 highlighted the importance of structuring songs with loops and adjusting them live, creating a cohesive song structure within the interface. P4 described adding various elements like orchestral pieces or voice tracks to a basic structure, using tools like SoundCloud and audio equalizers for refinement. P5 emphasized the common technique of using loops for creating transitions, while also experimenting with alternative methods like scratch transitions. P6 noted using loops to create specific effects such as hi-hats, and P7 discussed the significance of adding small touches, such as orchestral additions, to impact the overall composition.

Musicians construct music channel by channel: Most participants(6/7) share that Music production often involves building tracks layer by layer, adding different instruments and elements to create the full composition. There can be two sub-themes, layering process and structuring the songs. By layering, Producers typically start with a base and then add different instruments, building the track up gradually. They may experiment with bringing in elements from previous tracks to see if they fit into the new creation. Secondly, Musicians create song structures within their digital interfaces, organizing sections like intro, verse, chorus, and so on. This process helps in arranging the various elements of the composition coherently. P1 mentioned building tracks layer by layer, starting with a base and adding various instruments. P2 elaborated on the ability to record multiple loops from different sources and switch them on and off as needed. P3 highlighted the importance of structuring songs with loops and adjusting them live, creating a cohesive song structure within the interface. P4 described adding various elements like orchestral pieces or voice tracks to a basic structure, using tools like SoundCloud, some instrument sounds, and audio equalizers for refinement. P5 emphasized the common technique of using loops for creating transitions, while also experimenting with alternative methods like scratch transitions. P6 noted using loops to create specific effects such as hi-hats, and P7 discussed the significance of adding small touches, such as use a synthesizer to create an orchestral piece and then additions of another one, to impact the overall composition.

Effects addition to Each Channel: All Participants (7/7) stated producers experiment with

various effects to shape and enhance the sound of individual channels or instruments, which can dramatically alter the overall feel of the track. P1 discussed experimenting with effects to see how they change the sound, noting that the right effect can significantly alter a track's vibe. P2 talked about creatively using effects like reverb to enhance specific parts of a song. P3 did not explicitly mention effects but implied their importance in the broader context of music production. P4 emphasized that effects should be adjusted according to personal taste to achieve the desired sound. P5 highlighted that effects are essential for creating the right atmosphere in a track, particularly in using reverb to expand vocal sounds. P6 described applying effects to different instruments and channels using digital interfaces, and P7 mentioned experimenting with different effects on orchestral pieces to refine the sound. **Types of Effects:** Common effects include reverb, compression, and equalization. These can be used to make sounds grander, adjust frequency ranges, or clean up vocals. **Personalization:** The choice and application of effects are often based on personal taste and the specific needs of the track. Musicians may tweak parameters like threshold or compression to achieve their desired sound.

Visualization: Five out of seven participants (5/7) highlighted the role of visualization in applying and adjusting sound effects during production and noted that visualizing effects helps them understand their impact on the track, leading to more informed adjustments. P2 expressed the importance of visual indicators in managing live performances, particularly for identifying frequency ranges. P4 emphasized that visualizing frequency charts helps in determining the most prominent zones of an instrument, making it easier to balance the mix. P5 discussed how visual feedback on a mixing table, such as BPM displays, enhances the precision of adjustments. P6 noted that visual and physical feedback is crucial for making accurate sound adjustments, while P7 mentioned the importance of visual feedback in receiving and implementing external feedback during production.

Feedback and Sensitivity Six out of seven participants (6/7) stressed the importance of feedback and sensitivity in the iterative process of music production. P1 highlighted the significance of system feedback in adjusting controls, using a trial-and-error approach to refine the final sound. P4 discussed how visual and auditory feedback are essential for making precise adjustments, particularly when dealing with virtual instruments and effects. P5 and P6 both noted the importance of physical feedback when playing instruments, indicating that this feedback loop is crucial for maintaining control over the production process. P7 also highlighted the benefits of digital interfaces, which provide a high level of control and are essential for experimenting with effects on the fly and refining tracks based on feedback from others.

Learning Curve and Customization: Three of the participants (3/7) talked about the learning curve and customization of their systems. Multiple participants color-coded their channels and organized them in groups to play music. Some participants also suggested having the possibility to use gestures rather than clicks, as it's more intuitive to perform many actions.

5.2 Design Implications

Based on the results of the themes obtained from Thematic Analysis, we identified four main problems in channel selection, loop systems, adjusting effects, visualization, feedback, and controls sensitivity.

The Design Implications are as follows:

D1: Systems should allow hands-free gestures to create and modify objects like tracks, effects, and loops

The design implication arises from the theme of the learning curve and system customization. Tangible and haptic systems present challenges, such as limited flexibility, increased complexity, and a steeper learning curve. Physical controls tend to be less intuitive and more difficult to customize, making it challenging to adapt to new or evolving use cases. These limitations can be mitigated through the implementation of hands-free gestures, which facilitate easier and more fluid movement, thereby reducing reliance on physical controls.

D2: System should provide support for gestures that users create along with predefined gestures to support the movement, which also means helping the system be self-adaptive, rather than learning the system every time

Multiple participants mentioned the need for customization and learning of new systems, which makes it essential to offer adaptive interfaces. An adaptive system is a system that learns and adjusts to users' behaviors and preferences over time, providing a personalized and efficient interaction experience. These systems should allow users to customize their interactions, have preferences, and adjusting to the user's style over time, which would improve efficiency and help reduce the learning curve.

D3: System should facilitate intuitive layering and structuring of sound elements to support gradual build of Composition

Most of the participants believe systems should offer features that allow seamless addition, modification, and organization of layers and tracks, enabling musicians to explore more possibilities and maintain coherence. A composition typically consists of an introduction, a body, and a conclusion (outro). A gradual buildup in a composition refers to the process of taking small sound elements and layering them or incorporating them incrementally throughout the introduction, body, or conclusion. This technique involves the repeated placement and integration of these elements to construct and develop different sections of the composition.

D4: System should allow users to perform operations across channels and support customization smoothly

Three of the participants talked about the learning curve and customization of their systems. While current systems allow working across channels, the interactions aren't always easy and simple to learn. By developing simpler and more user-friendly interactions for performing operations across channels, there is potential to enhance customization options for these channels, thereby reducing the resistance to learning and adopting the system.

D5: Systems should enhance visualization capabilities to support real-time adjustment and fine precise control over sound elements

Visualization was a key theme, with participants stressing for the need for visual feedback to understand and adjust sound effects accurately. Sounds should incorporate features like BPM (Beats Per Minute) indicators, waveform display along with separation of Audio elements like Vocals, Instruments, and Bass, and frequency charts to provide clear real time visual feedback. It empowers users to make informed decisions during the production. Physical tools, such as knobs, sliders, and touchscreens, offer tactile feedback that enables precise adjustments through direct physical interaction, which is well-suited for tasks requiring exact control. However, gestures provide a more fluid and natural interaction, allowing for intuitive and immersive control over sound elements. It is ideal where natural movement enhances the creative process.

The figure shows the brainstorming process that assisted in coming up with design theme.



Figure 5.1: Brainstorming based on themes

Prototype Design and Implementation

Based on the study we come up with three *main research questions*:

1. How can musicians use digital tools to create rhythms?
2. How can you design a system to create rhythms digitally?
3. How do you build such systems?

RQ1: How can musicians use digital tools to create rhythms

This question explores how musicians currently utilize digital tools for rhythm creation, as observed in the study above. The findings and design implications derived from the study guide us to the next research question.

RQ2: How can you design a system to create rhythms digitally

This question examines existing tasks related to rhythm creation and the interaction techniques used to manipulate them, as detailed in Table 5.1 and Table 5.2. The design implications identified in the study are discussed, with a focus on how two of these implications inform the system design.

RQ3: How do you build the system

This question addresses the construction of the system, drawing on Human-Centered Design principles to inform both the design and technological aspects. The development begins with a web application, and the technical details of this process are covered in the sections below.

6.1 Design Concept

Based on the insights and open questions outlined above, a design concept was developed to address design implications D1 and D2. This concept comprises two main approaches. The first approach involves the use of fixed gestures that are mapped to the system. The second approach aims to empower users to create their own gestures and map them to the system. Various AI models are employed to facilitate gesture recognition, with the objective of assessing whether there are noticeable differences in user experience and system performance. The focus remains on specific interaction use cases within the context of music, particularly those that could offer significant value for digital music production.

Following a brainstorming session, several potential solutions were identified that might be relevant for future use cases.

1. **Gesture Based Interactions:** Simple and intuitive gestures for various music production elements, such as channels, loops and effects
2. **Channel Management and Layering:** Techniques for organizing and manipulating music layers and tracks to facilitate a seamless composition process.

3. AI Integration for Sound Manipulation: Incorporating AI to assist with tasks like gesture detection, explainable AI, and generating new audio elements.

4. Visualization and Feedback Mechanisms: Enhanced visual and auditory feedback to inform users of the effects and changes applied to their compositions.

5. Customization and Adaptability: Ensuring the system supports personalized workflows and adapts to individual user preferences.

After the ideation process, it was observed that significant challenges exist in designing systems that rely on gestures for operations requiring extensive music creation. Music creation tools often have complex interfaces that demand coordination with keyboards and mice, making it difficult to execute these tasks using gestures alone, particularly when it comes to moving items across a channel or multiple channels. To address these challenges and reduce the complexity and learning curve, a design concept was developed that incorporates a limited set of gestures specifically designed to select and move elements within the interface. The table below exemplifies these simple gestures:

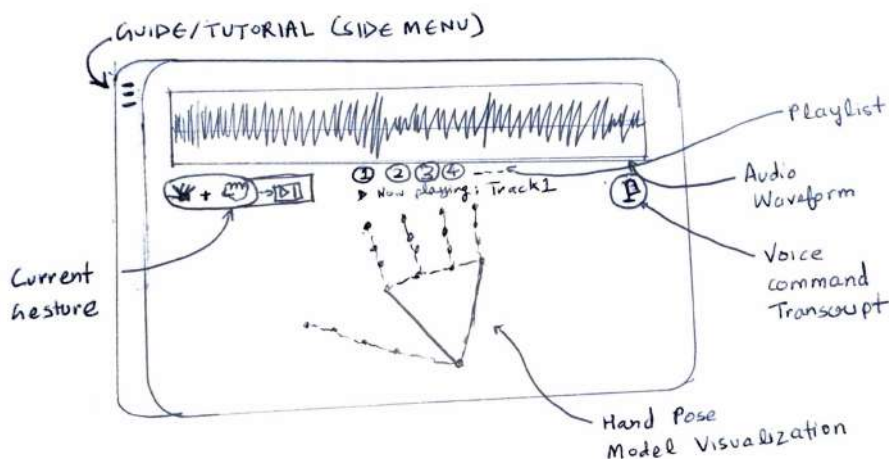


Figure 6.1: Sketch of Main Prototype

Task	Gesture
Select Item	Hold thumb and index finger together and move around
Move Item	Point finger to select and move

Table 6.1: Hand Gestures for Selecting and Moving Items in 3D Space

Figure 6.1 shows the design of the main user interface for the system. Figure 6.2 shows the multi-channel interface, which lets you add and edit multiple channels. It was developed based on ideas from brainstorming sessions. Figure 6.3 shows a storyboard sketch that demonstrates how to create, record and manipulate live music elements. It's also based on the design implications and features that were discussed during previous brainstorming sessions.

Figure 6.3 shows how the storyboard progresses from left to right, starting with the first and second rows. The user first selects a loop using gestures, then copies it with another gesture.

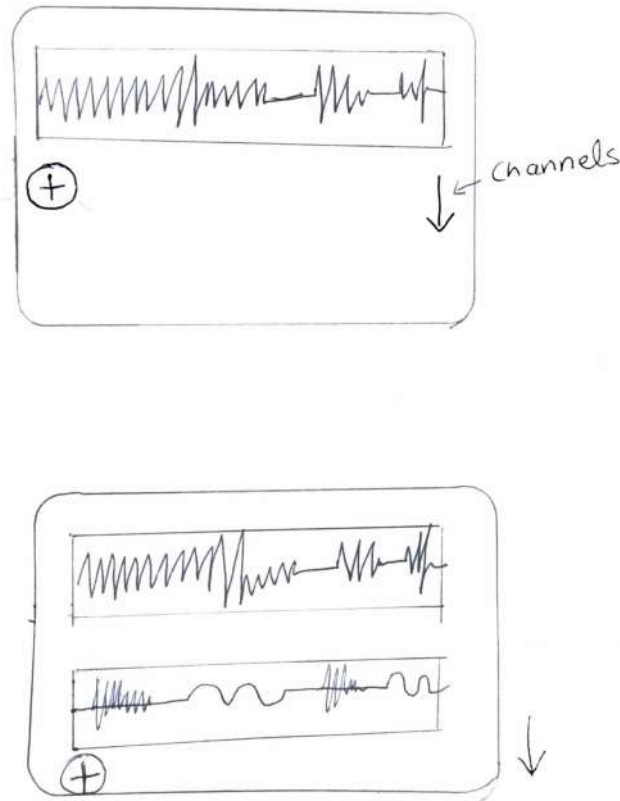


Figure 6.2: Sketch of Multi Channel System

They then perform drum patterns using gestures, and finally record vocals with the help of an external visual plugin for guitar bands to complete the track. The storyboard shows a hands-free interaction model, which lets users perform all actions through gestures.

It was observed that, when utilizing a music manipulation application, certain features were more crucial than others for the system to function effectively on various mobile devices, including smartphones, tablets and other portable devices. [22] As a result, a number of essential features for the DJ system have been identified, as shown in Table 6.2. In addition, Tables 6.3 and 6.4 highlight the different interaction techniques used in professional DJ equipment and digital audio workstations (DAWs), respectively.

Below is a look into the different ways DJs interact with their systems.

Further, we also take a look at how music producers look at using digital tools to work with music.

6.2 Implementation

We developed a web application leveraging modern UI toolkits and Google Media Pipe for crafting tunes. The implementation covers technical aspects including frontend and backend development. We utilized Computer Vision to track finger movements and mapped musical instruments to gestures pre-defined and user-defined both.



Figure 6.3: Storyboard Sketch explaining how various interactions can help in Loop, Copy and other operations

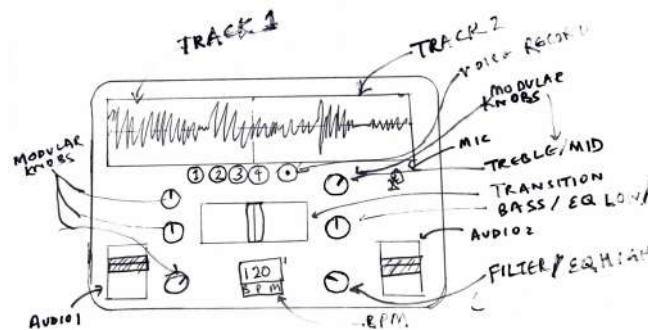


Figure 6.4: Sketch of Prototype for Future



Figure 6.5: Main interface of the system

Category	Details
EQ and Filter	Bass: Adjust the low-frequency sounds.
	Mid: Adjust the mid-frequency sounds.
	High: Adjust the high-frequency sounds.
	Filter: Apply a low-pass or high-pass filter to modify the track's frequency spectrum.
FX (Effects)	Echo: Repeats the sound with diminishing volume, creating an echo effect.
	Reverb: Adds reverberation, making the sound echo as if in a large space.
	Kick: Adds a kick drum sound to the track.
	Looper: Repeats a section of the track in a loop.
Sampler	Airhorn: A loud, attention-grabbing sound often used in DJ performances.
	Siren: A continuous, high-pitched sound used to build tension or excitement.
	DJ Drop: Pre-recorded vocal samples (e.g., "DJ, drop that beat!") used to personalize sets.
	Scratch: A sample that mimics the sound of vinyl scratching.
Sync, Hot Cues, Jump	Sync: Automatically matches the BPM of both tracks for seamless mixing.
	Hot Cues:
	Cue 1
	Cue 2
	Jump:
	Jump -8 (jumps back 8 beats)
	Jump +8 (jumps forward 8 beats)

Table 6.2: Essential items for a DJ system.

Key Interface:

- Main Canvas: Central Workspace where musicians can arrange and manipulate tracks, loops and other effects
- Guide: A mapping of each gesture to the command/music operation
- Feedback Icons: Displays real time visual feedback, such as waveform display to assist in precise sound manipulation
- Gesture Input and Feedback Panel: A dedicated area for creating, editing and applying gestures to different music elements, also seeing the gesture and your hands
- Voice Component: Transcribe live audio to commands and text display
- Music Visualization Waveform: Displays the waveform of music playback

Interface Implementation for Fixed Gestures*I. MainComponent.tsx*

Task	Interaction Style
Blend/Mix Tracks	Use Crossfader
Adjust Tempo/BPM	Move Pitch Slider
Sync Tracks	Press Sync Button
Create Loops	Press Loop Function Buttons
Copy & Loop	Use Hot Cue Buttons
Track Selection	Rotate Jog Wheel or spin Platter
Apply Effects	Turn Effect Knobs/Buttons
Visual Feedback	View Waveform Display
Beatmatching	Use LED Indicators/Jog Wheel Lighting
Track Progression	View Waveform Display
Drag & Drop Samples	Use Touchscreen Interface

Table 6.3: Tasks and Interaction Styles Supported by Pioneer CDJ-3000 with Rekordbox

Task	Interaction Technique
Write melodies and arrange song structure	Click and Type
Create and modify loops	Drag and Drop loops into timeline
Adjust tonality, tempo, and volume	Click and Drag sliders/knobs
Copy, loop, and arrange channels	Right-click and Select (e.g., copy, paste)
Manipulate channels (e.g., isolate, accelerate)	Click and Drag
Apply effects like EQ, reverb, and delay	Turn Virtual Knobs or Click and Drag
Reflect on and iterate ideas	Click and Review (past iterations)
Real-time visual feedback on sound	Real-time Display with Click to Adjust
Drag and drop tracks and effects	Drag and Drop

Table 6.4: Important tasks and interaction techniques in music creation using FL Studio, Ableton Live, and Logic Pro X.

Interaction Technique	Main Feature
Right Hand (open hand) + (fist)	Play/Pause
Right Hand (thumbs up) + Rotate	Control speed
Right Hand (pointing up) + (left-right arrow)	Volume control
Left Hand (open hand) + (OK sign) with every finger	Play the drum
Left Hand (peace sign) + (crossed fingers)	Start a loop
Right Hand (peace sign) + (crossed fingers)	Close a loop
Left Hand (peace sign)	To remove a loop

Table 6.5: Gestures and their corresponding main features.

The main application component, responsible for rendering and coordinating other components.

Interactions:

1. Webcam: Initialized on component mount if supported by the browser.

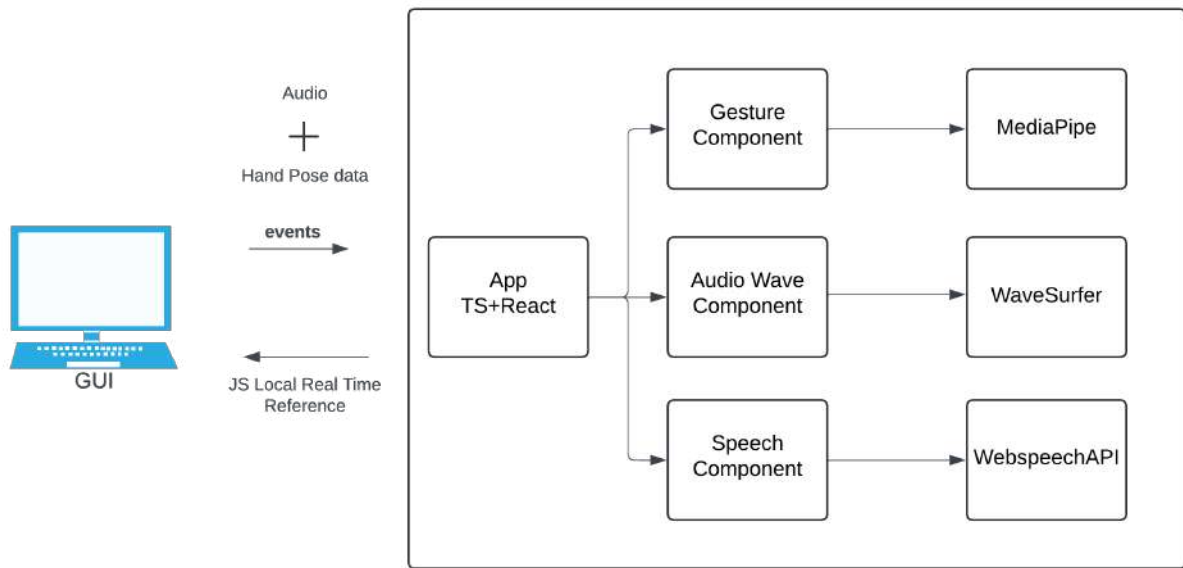


Figure 6.6: System Architecture Overview

2. Audio: **AudioWaveComponent** renders and interacts with the audio waveform.
3. Gestures and Speech: **GestureComponent** and **SpeechComponent** interact with the waveform, webcam stream, and mic to provide functionalities

II. *AudioWaveComponent.tsx*

Component responsible for visualizing audio waveforms. It utilizes the **WaveSurfer** utility from the **wavesurfer-react** library. It is responsible for rendering an audio waveform visualization of the audio file. Controlled and interacted with from its parent component.

III. *VolumeProgressBar.tsx*

Displays a progress bar indicating the volume or amplitude of the audio. It provides visual feedback on the volume level to the user. This is achieved using a progress bar where the width represents the volume percentage.

IV. *GestureComponent.tsx*

Provides intuitive and dynamic control over audio and video playback through hand gestures and voice commands. It interfaces with the **GestureRecognizer** from the **@mediapipe/tasks-vision** library to recognize gestures, and it uses the results to control various functionalities like audio playback, volume adjustments, and special effects.

Dependencies:

1. *React*: Utilize React for building user interfaces efficiently.
2. *WaveSurfer.js*: WaveSurfer.js is used for audio waveform visualization and playback control.

3. *MediaPipe*: MediaPipe is used for real-time hand gesture recognition.
4. *Bootstrap*: Bootstrap CSS is employed for styling and layout purposes.

Key Functionality:

1. Real-time hand gesture recognition
2. Audio playback control
3. Looping and region creation
4. Volume adjustment
5. Effects control for playback speed

Enums are defined to represent the states and actions that the module can take: CutState, PlayPauseState, IndexState, MiddleState, RingState, PinkyState, VolumeState, and EffectsState.

V. GestureModel.tsx

Defines the data model for gestures and their associated labels. This is the data that the gesture component uses.

VI. SpeechComponent.tsx

Handles the recognition of voice and performs actions based on the detected words.

VII. AudioManager.tsx

Loads the audio files to permit their playback during web code execution. Initializes the audio context for each sound, storing it in a buffer for recall as needed.

Further, as the discussion moves towards the second scenario: User Defined Gestures. What methodologies might be employed to utilise machine learning in the generation of musical rhythms, and how might the resulting outputs be made comprehensible? This study explores the potential of explainable machine learning in the context of music production.

The methodology employed utilises supervised machine learning models, which are trained on labelled data. Each training data set is associated with a designated output label or target. The objective is to predict the output for new, previously unseen data, or in this case, the labels for the gestures. The most prevalent supervised learning tasks are those of classification. This enables us to generalise from the training data to unseen situations in a manner that is both reasonable and accurate. We will discuss Algebraic Machine Learning and Neural Network and compare the results.

Machine Learning for User Defined Gestures

Our implementation takes place in three stages : The first is the user interface implemented using TypeScript and the web framework that handles data collection and processing developed using the same frontend system, as before. The second part is the AML, which generates trained AML models from the training examples provided by the user. Our AML server was implemented

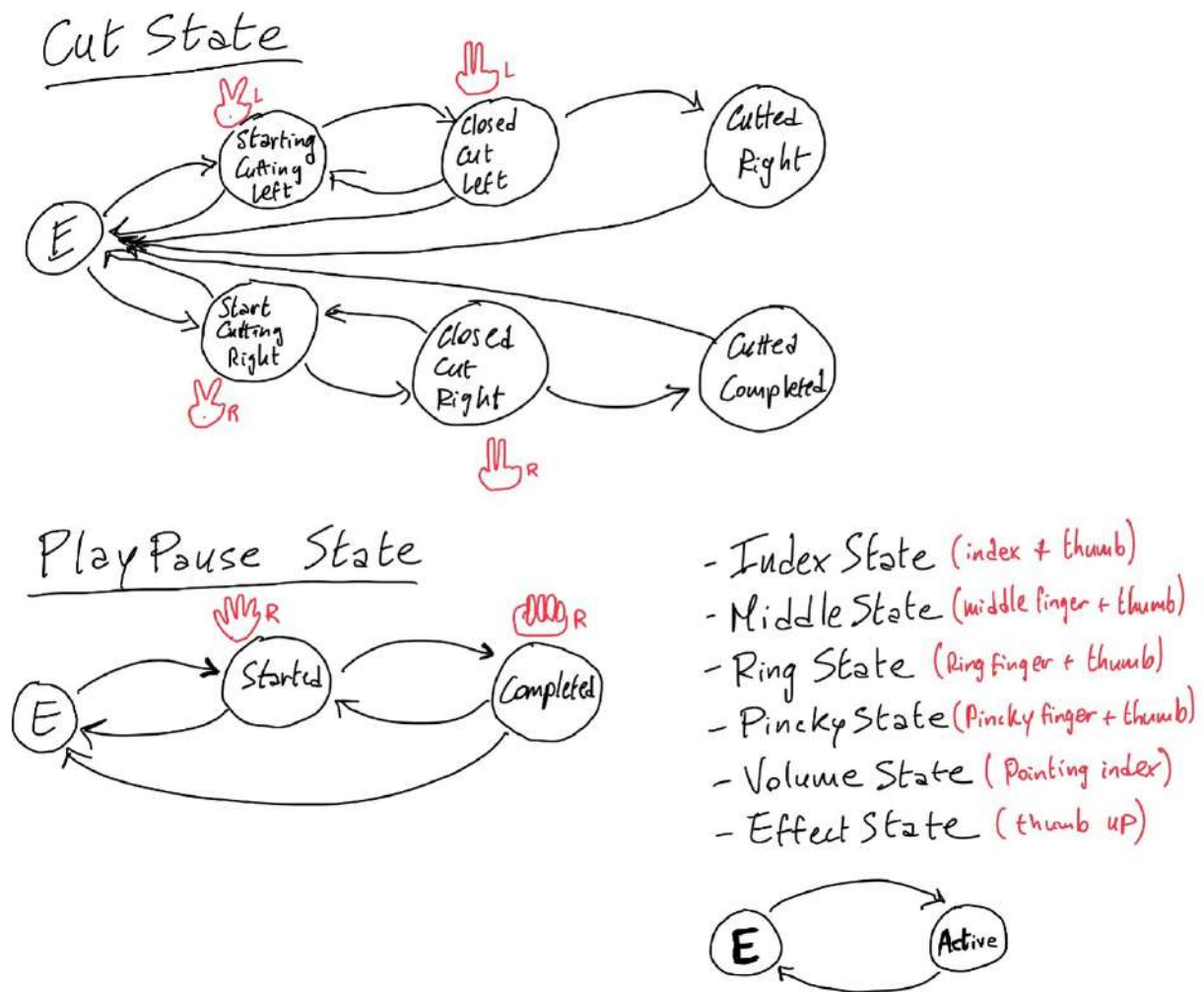


Figure 6.7: Finite state machines architecture: it's utilized to enable seamless transitions between various gesture states

in Python using Flask, as the AML engine has Python bindings. The model structure is then serialized to JSON and sent to the web application. For the comparison algorithms we used tensorflow-js, whose models can be easily integrated into our codebase, allowing the user to compare AML results with a large number of available models.

In the third part, the system maps personally defined gestures to sound patterns using real-time prediction based on the selected model.

Data Collection & Model Discussion

Data collection happens using the user's camera. Images are fed to a hand pose estimation model [23] the user's hand joints in real time, running in the browser. We use the MediaPipe Hand Pose detection model available via Tensorflow [26]. For one hand, we use as features both the image coordinates and the 3D position of the index finger. As we are interested in gestures,

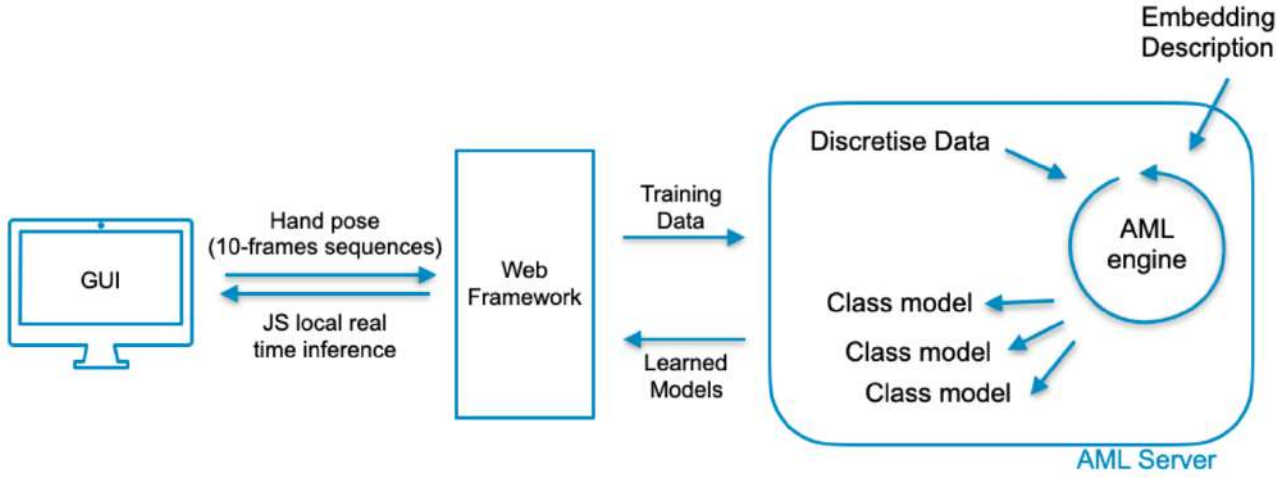


Figure 6.8: AML training pipeline with three stages: The interface (GUI) and Web framework in which we collect and process the gesture input data; The AML server including the AML engine to build the model classes; and The real time prediction based on the learned models.

which means movement, each instance is produced based on a 10-frame sequence of hand pose coordinates. That is, once the user starts recording data for a specific gesture, every sliding window of 10 frames will generate an instance. In other words, we use a window of size 10 and slide 1 which creates instances as long as the data collection process is active.

Training data for AML

For our 10-frame window, at each step we gather 5 values. The first two represent the x and y image coordinates of the index finger, respectively. The last three represent the x, y and z coordinates of the index finger in the hand coordinate system as extracted by [23]. This means that each example is a vector of size 50 where the first 5 numbers correspond to those values for the hand pose detected at the first frame of the sequence and the second 5 values to those of the second frame and so on. If the hand is not detected at a specific frame of the sequence, the 5 values corresponding to it are assigned zero.

AML Server

For our 10-frame window, at each step we gather 5 values. The first two represent the x and y image coordinates of the index finger, respectively. The last three represent the x, y and z coordinates of the index finger in the hand coordinate system as extracted by [23]. This means that each example is a vector of size 50 where the first 5 numbers correspond to those values for the hand pose detected at the first frame of the sequence and the second 5 values to those of the second frame and so on. If the hand is not detected at a specific frame of the sequence, the 5 values corresponding to it are assigned zero.

Discretize data

In its current form, our demo uses a version of the engine that can only deal with discrete data. Thus, in the AML Server, each feature is discretized individually, using [25] to obtain for each feature the optimal number of bins from the training data. The bin edges will also be returned later to the Web Framework together with the learned class models, as it is needed for real time prediction in the browser. Later, we will show an example of a single feature being discretized for a sample dataset with 100 examples. It will illustrate how we use the distribution of the first 3D 'x' finger coordinate of the gesture to discretize the feature into a reasonable number of bins using a bayesian approach.

In the next few sections, it will involve the mathematics behind the model, and more context for better understanding. The research involves work of the authors, the researchers at DFKI and Janin Koch. There is also a section on embedding function in the appendix. [45]

AML Models

As explained previously, we train one model per user-defined class. Training in the AML sense means producing atoms, which are sets of constants related to each class. Thus, each class model will have a (potentially different) number of atoms.

Once training finishes, the AML server sends back a JSON representation of the atoms for each class model together with the description of discretization bins, so that we can convert every real value in real time to its bin, then obtain the algebraic constants related to that bin. From the union of all instance constants we can calculate the number of atoms that are present for each class and from the number of missing atoms provide a class prediction. In the ideal case, a positive sample of a class should have zero atoms missing for it. In practice, we provide classification outputs and estimate confidence using the cumulative number of misses observed in the negative examples of the training set for each class. In more formal terms, the score a test example x has for class k is

$$1.0 - P(\geq m_k \mid -)$$

where m_k is the observed number of misses for x according to the model where k is the positive class and P is the probability of misses computed only over negative training examples of class k . Confidence values are then computed by normalizing confidence scores across classes.

NN Models

For the neural network we allow the user to specify not only the number of training iterations (epochs), but also the hidden layer size. This is not needed for AML models, as the training procedure will pick the representation size (number of atoms) for each class. We also provide for the neural network with a progress report on the loss and train / validation performance. At this point we are not providing anything equivalent for AML, as this is included in future work. AML has no training error for all seen training examples (their number of misses should be zero for their ground truth class) and therefore this metric is not relevant. Calculating performance in the validation set would require computing the distribution of misses, which could increase considerably the training time and, moreover, computing and displaying those metrics would add further synchronization delays. We are still exploring how to provide better user feedback beyond classical machine learning performance metrics.

Real Time Prediction

Learned models

Users can explore the performance of both AML and neural network models. This can be done by inspecting the confusion matrix of each model in the recorded dataset. We plan to apply the standard confusion matrix function to estimate these numbers.

Real time inference

The live prediction of the trained hand gesture is based on the camera stream, whose values are sent to the hand detection pipeline and then to the trained models, which are loaded in the browser in our training step and run locally. In the case of AML, this means discretizing the obtained data on the fly based on the bins for each feature, obtaining the constants associated with those bins, and determining the number of atoms present for each class from the union of all instance constants. This is all implemented in the browser in Javascript using set operations (as an atom is considered to be present in an instance if one of the atom's constants is present in the example). From the number of missing atoms, we can predict the class as explained above.

The confidence level is then mapped to the volume of a pre-determined audio track, which is played in loops to allow a smooth continuous stream of sound to be explored. The system can be retrained easily for different instruments or to change the gesture, it is possible.

6.3 Scenario

The following scenario illustrates how Max, a DJ and musician would use Tune Crafter to play music.

Track Selection and Playback

Max begins by speaking the name of the track he wants to play. The voice detection system picks up the track. He then selects multiple tracks and eventually settles on a techno track. Max initiates playback by performing a closed fist action, which the system recognizes to start the track.



Figure 6.9: *

(a) Track selection and playback initiation via voice detection and gesture.

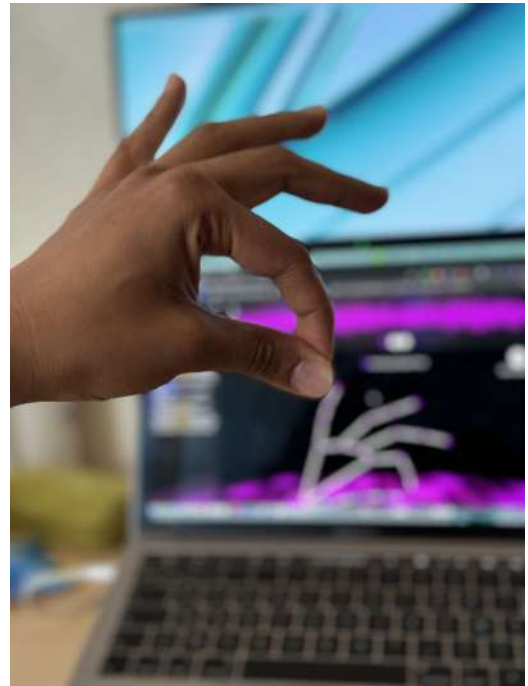


Figure 6.10: *

(b) Creating drum beats using finger taps.



Figure 6.11: *

(c) Transitioning to the next track by monitoring waveform density.

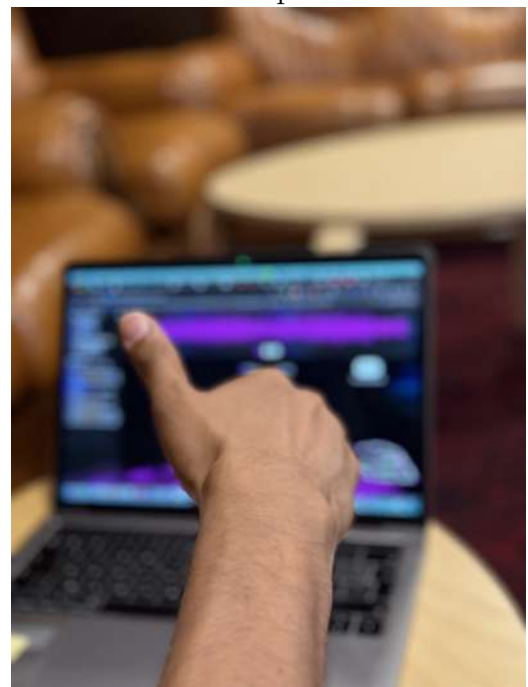


Figure 6.12: *

(d) Adjusting speed and mixing effects using thumb gestures.

Figure 6.13: Gestures used for various actions in the music creation process.

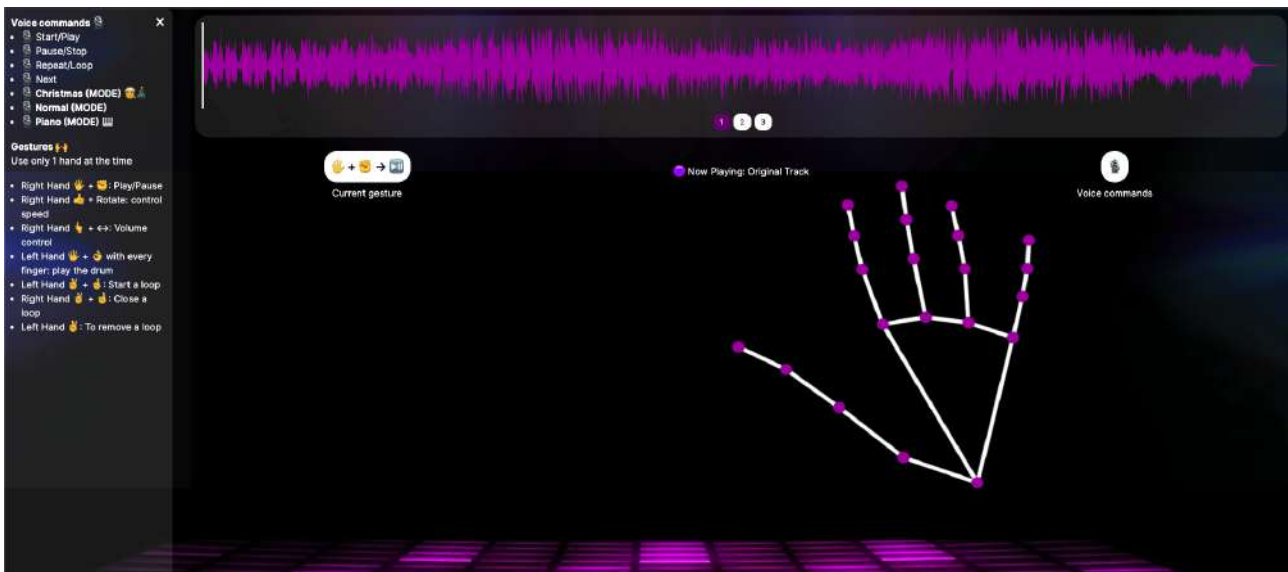


Figure 6.14: Track selection and playback initiation via voice detection and gesture.

Drum Beat Creation

Max selects the drum module next. He plays the beats in order by tapping his fingers on the surface, and the system captures his input accurately.



Figure 6.15: Creating drum beats using finger taps.

Track Transition

After performing the desired beats, Max decides to transition to the next track. He waits for the waveform to show a less dense area, which likely indicates the outro or breakdown where the energy decreases. Max either says the name of the next track to transition or clicks on the next track button.

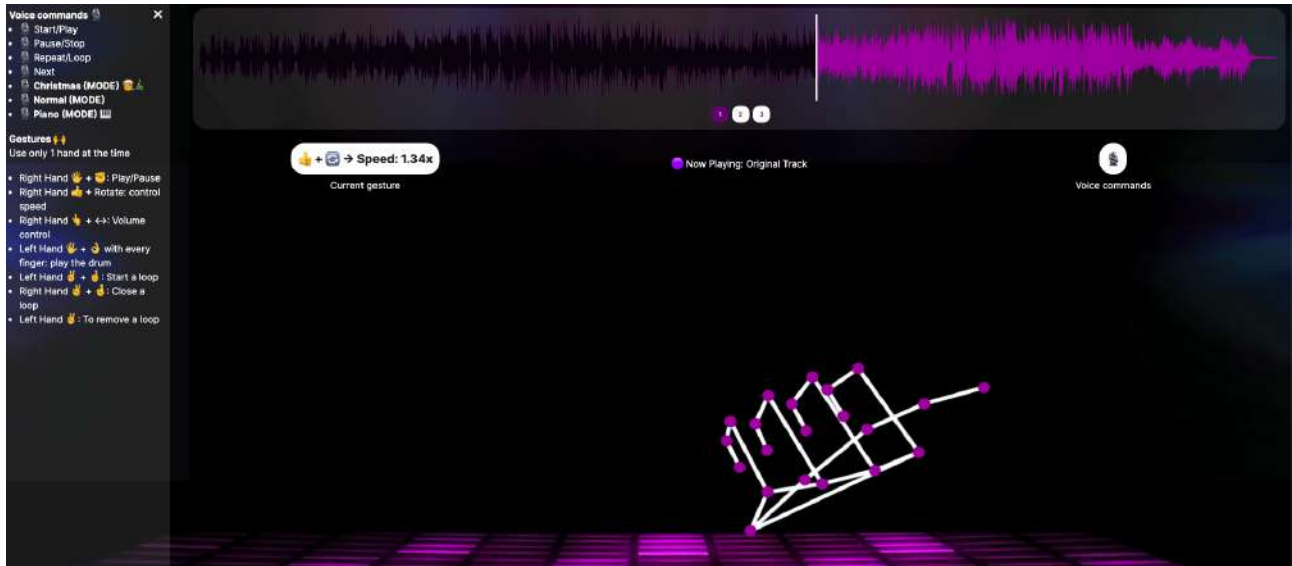


Figure 6.16: Transitioning to the next track by monitoring waveform density.

Speed Adjustment and Effect Manipulation

Max then decides to change the speed of the music. He rotates his thumb from the center to the right, moving the speed from 1x to 1.5x. As he desires a slow effect, he mixes it by moving his thumb around to create different effects and finally returns to the normal speed.

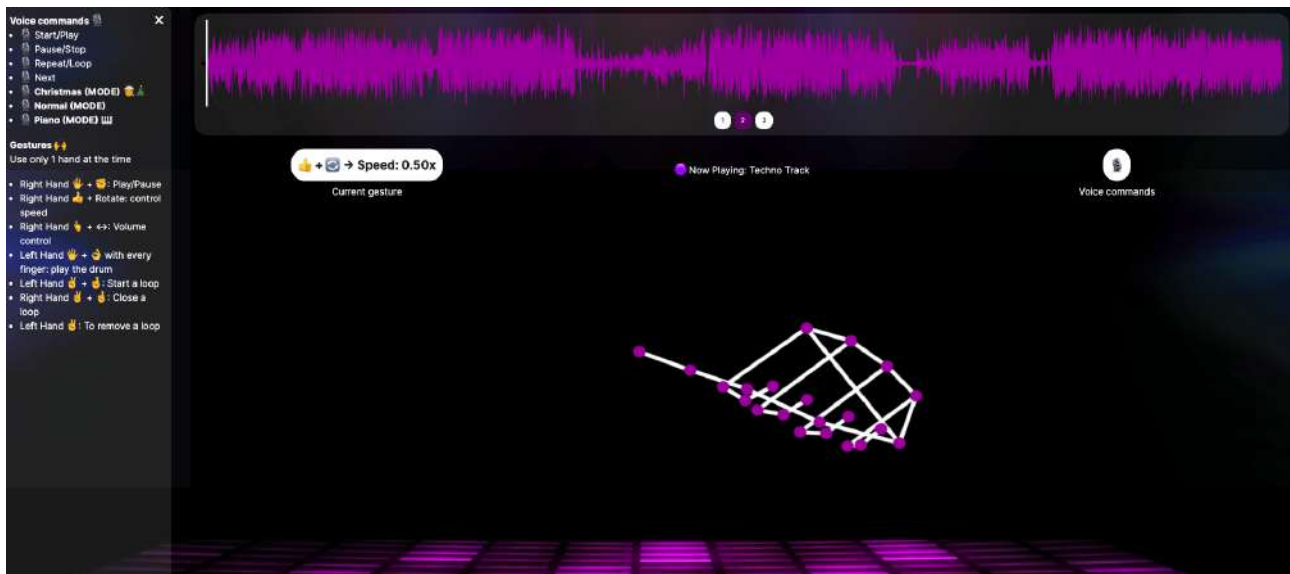


Figure 6.17: Adjusting speed and mixing effects using thumb gestures.

User scenario for Self Created Gestures

Further, Max also wishes to experiment with designing his own gesture so 1. he types the effect he wants to perform and then performs the gesture. 2. After recording multiple gestures linked to corresponding effects, he starts the train of gestures. 3. Further, after the training is complete, he can perform the gesture to see when he performs the gesture again, and how the effect is being recognized to get the right effects.



Figure 6.18: Set the label and start the gesture recording

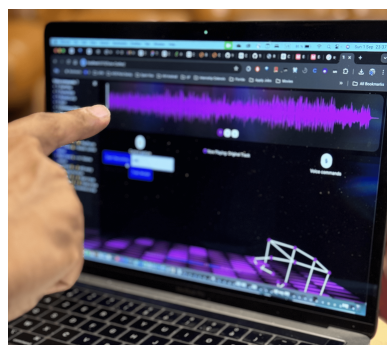


Figure 6.19: Midway through the gesture

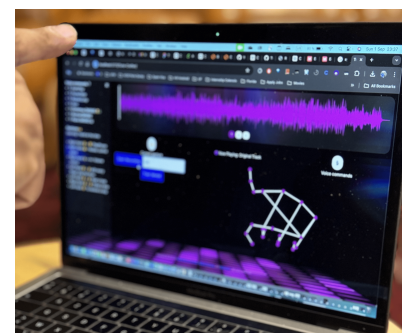


Figure 6.20: Gesture recorded

Figure 6.21: User created gesture recording scenario

if he identifies a low accuracy he can go back to add more training data and/or rerun the model to improve the prediction quality. When he starts the video and starts the real time

prediction of the models, he can see the effects of the music based on the predictions. Afterwards he can start the prediction for the comparison model and observes how both models compare the gestures. After going through the process multiple times, he gains better understanding of how the model trains and what the limitations of each model are.

Discussions

7.1 Choice between Gesture Interaction and Other Methods

One big portion of designing a new way of interaction is whether the gestures work for the user. If you look at the paper on ubiquitous computing, the augmented reality headsets are working well with gestures. As intuitive interactions will replace traditional interactions with devices, we have an opportunity here. Our work tries to fill the gap for research in interactive systems in music digital tools in the space of production. Evaluations should help understand whether the interactions are much more useful to people or whether they are averse to changes. Many of the participants remarked that they like the traditional systems because they have more control and the feeling of tangible devices are hard to replace along with the ability to fine tune with precision. We believe a study on the same would be very remarkable and useful. [18]

7.2 Limitations

This study was subject to a number of limitations, which will be discussed in turn. Firstly, the user study was constrained by time, resulting in the recruitment of only seven participants, all of whom were male. A more diverse cohort, including participants of different genders, ages, and regions, would have enhanced the generalisability and applicability of the findings. Secondly, when proposing FX and samples, there are intrinsic limitations regarding the number of samples that can be incorporated into the tool and their applicability to different audiences. Given that audience members frequently hail from disparate backgrounds, including marketing, cinema, or commercial music, it is inevitable that not every sample will be pertinent or appropriate for all users. Thirdly, whilst hands-free gestures may facilitate easier movement and interaction, they may also result in user fatigue, particularly in scenarios that require prolonged use. This is in accordance with research findings on mid-air gestures. [18] Additionally, the system, which records and recognizes live data, operates relatively slowly and could benefit from further optimization. If the module for storing and tracking gestures were improved, efficiency could be increased. Currently, the system exhibits slower performance and detection issues, particularly when both hands are used, which represents a challenge that we aim to address in future iterations. Transitioning between different tracks or channels remains difficult without these improvements. Furthermore, there is a slight delay between gesture detection and sound playback, which also requires reduction.

7.3 Future Work

Further studies are required to gain a deeper understanding of the subject matter. It is our contention that the design concepts that emerged from the various design implications have significant potential and will be implemented in the future. Moreover, there are numerous research projects, such as the work on Mid Air Zoom [18], that can be extended or tested with

the integration of multi-gesture systems. The alternate design concepts are also conceptualized to be implemented in the future. Knotty Gestures allow users to interact with handwritten marks, turning them into functional elements, such as linking symbols to audio recordings or assigning meaning to underlines and characters. An implication is we can extend the work and try to incorporate gestures to store information and access it securely. [19] The work conducted by Open AI's Jukebox and Google Music FX are open-source music projects that facilitate the creation of AI-generated music. [21] [20] The field of generative AI music is relatively new and has not been extensively studied. However, there is significant potential for research and the implementation of ML models with innovative interfaces, as evidenced by the numerous academic papers on the subject. In the timeline of the project, there is an evaluation study planned around the mid-September.

Conclusion

The increase in digital tools in music production presents both opportunities and challenges for musicians. The primary challenge lies in the complexity of these systems, which often hinders intuitive interaction, particularly in creative workflows where spontaneity and fluidity are of paramount importance. This research project was designed to address three primary research questions.

1. How can musicians use digital tools to create rhythms?
2. How can a system be designed to create rhythms digitally?
3. How can such systems be built?

Through a qualitative study and design iterations, several critical insights into how musicians approach digital music production were identified. These include the importance of layering sounds, real-time feedback, and the necessity of maintaining control over individual channels. These insights informed the design and development of Tune Crafter, a prototype system that integrates gesture-based interaction and explainable AI, offering both pre-defined gestures and user-defined gestures. The system is intended to facilitate the integration of musicians' creative needs with the functionality of modern digital tools.

This work makes three key contributions to the field. Firstly, an empirical study was conducted which illuminated the key challenges faced by musicians when working with digital tools. This study provided the foundation for the design decisions that were subsequently made. Secondly, a design theory methodology was applied in order to create a prototype that enhances interaction through gesture control, thereby allowing for greater flexibility and creativity. Thirdly, the potential of Live gesture detection and music interaction along with use case of Algebraic Machine Learning (AML) and other AI models was demonstrated in supporting real-time, explainable interaction, thereby enabling musicians to train and customize the system according to their needs.

By bridging the gap between human creativity and digital tools, the work strives to empower musicians to transcend the boundaries of musical expression, making digital music production more accessible, adaptable, and intuitive.

Acknowledgments

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Appendix : Interview Questionnaire

This section presents the interview questionnaire employed in the research study. The following questions were posed to participants during the semi-structured interviews.

1. Can you describe your workflow for creating a rhythm?
2. How do you explore new sounds and rhythmic patterns? How do you decide which new sound to integrate?
3. After you created an existing rhythm, can you tell me how you go about modifying it? What tools or techniques do you use to make adjustments?
4. How do you feel about using digital interfaces that mimic physical interactions, such as virtual drum pads or touch-sensitive surfaces? How do these compare to traditional physical instruments?
5. What types of feedback (visual, auditory, haptic) do you find most useful? How do you incorporate feedback into your rhythm creation process?
6. How does visualization of rhythms (waveform, beat grids) impact your workflow? Are there any specific visualizations you find particularly helpful?
7. Do you feel that your physical movements influence your creativity and musical expression? How so?
8. What types of movements do you find most natural or effective when creating rhythms? Are there any specific movements that you find particularly challenging or frustrating?
9. What are the tasks you perform every time you create rhythms? Are there activities that you do less frequently? Why?
10. How do you customize your workflow or tools when creating music? What aspects do you customize most often and why?

Demographics:

1. Age
2. Self-identified Gender
3. Background in Music
4. Type of Musical Instrument
5. Current Job Title
6. Years of Experience in Music

Appendix : Informed Consent Form



Informed Consent Form

Working Title

Exploring Machine Learning Explainability for Music Creation

Research Team

The study is conducted by Shubhankar (Masters Student), under the supervision of Janin Koch (Inria Permanent Researcher). All investigators work at the ex)situ team at Inria Saclay, which is lead by Prof. Mackay. Physical Address: B t. Ada Lovelace, Rue Raimond Castaing, 91190 Gif-sur-Yvette

Study Description

Study Description

In this study, we investigate the technical workflow musicians follow when creating and modifying rhythms. We explore how creative professionals interact with digital systems, focusing on the tools and techniques used in the rhythm creation process. Specifically, we aim to understand how different input modalities and the visualization of rhythms impact the workflow, and the role of customization in these tools. Through this study, we seek to uncover how musicians incorporate feedback, the importance of iterative refinement, and the potential for automation in rhythm creation. Our goal is to enhance existing systems and identify opportunities for improving the integration of AI in the musical design process, ultimately facilitating a more seamless and intuitive collaboration between musicians and technology.

Study Location

The study will take place in Paris Center or Saclay Plateau, upon mutual agreement between the experimenter and the participant. A stable internet connection is necessary.

Duration

The full study session will last approximately 30 minutes.

Figure B.1: Informed Consent Form

Data Collection, Confidentiality, Data Protection

The study will be audio and screen recorded. This includes recordings of screens and conversations during the study. Your participation in this research is confidential. The contact investigator, Shubhankar, will assign you a unique participant ID that will be associated with all of the data we collect. The investigator will maintain a separate list that links your participant ID to your name and other personal information, in alignment with the European General Data Protection Regulation (GDPR). Audio recordings will be transcribed or summarized in textual form by the investigators. The investigators will check all transcripts for any identifiable information, such as names, email addresses, locations, IP addresses, ID numbers, or any word and sentence that could reveal your identity. If any identifiable information is found in the transcribed data or the textual responses, it will be deleted. The scientific supervisor, Wendy Mackay, will store and secure the raw data on external storage devices in a locked office. They will be preserved for 15 years and will be destroyed after.

All publications and presentations resulting from this research will present pseudonymized quotations, pseudonymized examples of specific trials, or aggregate data, with no personal identifiers. In case the investigator wants to diverge from this norm and include any visual material from the interviews in any public available material, the investigators will contact you to request an explicit consent of use.

You have the right to ask the investigators to destroy your non-public data at any moment – the investigators will destroy them within a delay of 7 days from the day of your request.

Use of Data

The results of this research may be presented in scientific conferences and published in conference proceedings or a journal. The data are collected for sole research purpose and are not made accessible to third parties.

Voluntary Participation

Your participation in this study is entirely voluntary and you may stop at any time, without providing a reason and without penalty. You may also decide to avoid completing a task or answering a particular question without giving a reason and with no penalties.

Consent

Please contact shubhankar..singh@inria.fr, if you have any questions, complaints, suggestions or concerns about this research, or if you are interested in receiving a copy of a future publication based on this research.

If you agree to the information outlined above, please sign your name and indicate the date below. You will be given a copy of this consent form for your records.

The nature and purpose of this research have been sufficiently explained and I agree to participate in this study. I understand that I am free to withdraw at any time without incurring any penalty.

Participant ID: _

Participant's Name (please print)	Signature	Date
Investigator's Name	Signature	Date

Appendix : Additional Figures



Figure C.1: Plugins



Figure C.2: Dune



Figure C.3: Omnishpere



Figure C.4: Slicex EQ



Figure C.5: Valhalla Vintage Verb



Figure C.6: Dune 3



Figure C.7: Order 1: The first image showing the initial setup of making composition.



Figure C.8: Order 2: The second image showing the intermediate process.





Figure C.11: Order 5: The fifth image showing the intermediate process.



Figure C.12: Order 6: The last image showing the last step.

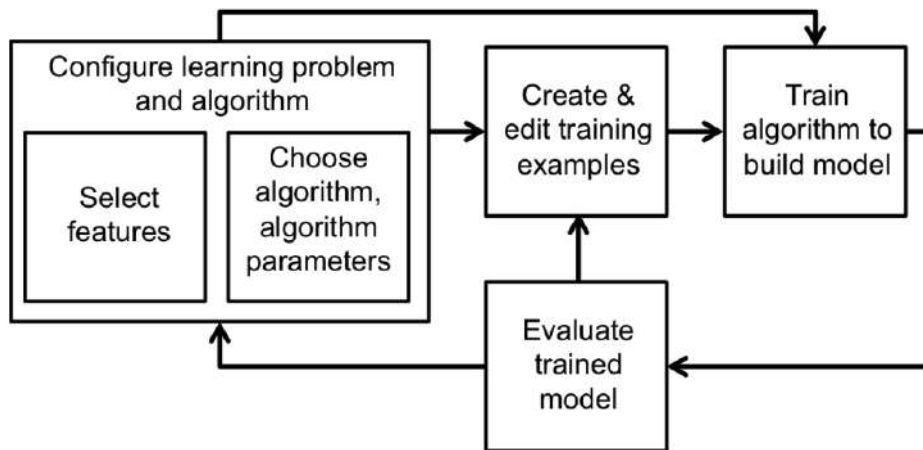
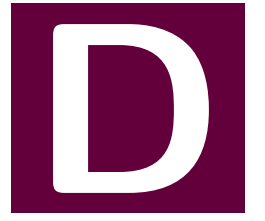


Figure C.13: ML Architecture [34]



Appendix : Theoretical Overview of the Algebraic Machine Learning (AML) Model & Glossary

This appendix provides a theoretical section of the Algebraic Machine Learning (AML) model, detailing its structure, the mathematics behind the model, and more context for better understanding. The research involves the authors, the researchers at DFKI and Janin Koch. [45]

Embedding description

In order to understand how AML represents our gesture recognition problem, one has to understand what in the AML terminology is called the *embedding*, that is, the representation of both the training data and domain knowledge. The overall structure of the embedding description is defined in the server before any user interaction takes place, but the exact number of constants it uses will depend on the number of bins obtained during discretization and the number of classes the user has created.

What is fixed is that our embedding assumes each feature is a discretized float number and therefore its bin number represents some ordered intensity. This notion of order for a feature f that has been discretized in nb bins is represented using $2(nb - 1)$ constants separated into two sets of $nb - 1$ constants each, one set for “lower or equal” constants and another set for “greater than” constants. Thus we have for each feature f constants $le_{0,f}, le_{1,f}, \dots, le_{nb-2,f}$ and $g_{0,f}, g_{1,f}, \dots, g_{nb-2,f}$.

The formal knowledge we want to express is the notion of intensity, which we will represent as ordered sets:

$$le_{i+1,f} \leq le_{i,f} \quad (1)$$

$$le_{i,f} \not\leq le_{i+1,f} \quad (2)$$

$$g_{i,f} \leq g_{i+1,f} \quad (3)$$

$$g_{i+1,f} \not\leq g_{i,f} \quad (4)$$

Forming one ascending chain and one descending chain for each feature. The AML server therefore has a Python function that generates those chains (constants and their relationship) in AML-DL based on the number of bins obtained for each feature in the training data. Other than the aforementioned constants, the AML embedding also defines a total of $2nb$ constraints linking the bin number to the label of a particular class in the training data.

Representing our input features, we also create one constant O_k for each of the n_c user-defined classes. Now we are ready to represent the training data in AML-DL. For each training example, which is a vector r , a term T was created representing the problem input:

$$T = \bigotimes_s \begin{cases} l_{eo,f} & \text{if } r(f) = 0 \\ l_{er(f),f} \circ g_{r(f)-1,f} & \text{if } 0 < r(f) < (n_b - 1) \\ g_{14,f} & \text{if } r(f) = n_b - 1 \end{cases}$$

and the training example is linked to its ground truth class with the positive relation:

$$O_k \leq T \quad (6)$$

and $n_c - 1$ negative relations expressing that this example does not belong to the other classes, meaning:

$$O_{k' \neq k} \not\leq T. \quad (7)$$

After the embedding is created, the server will, for each class present in the training set, run the training loop of the engine for the number of iterations specified for the user. Here by iteration we mean batch, as, similar to the way other machine learning methods operate, AML also processes data in batches. Each batch has a number of positive examples sampled from the training dataset and a number of negative examples also sampled from said dataset. The initial number of positive examples in a batch is fixed as 50, while the initial number of negative examples is set as 100. Those numbers change in subsequent iterations batches using a heuristic provided by Algebraic that takes into account the false positive rate and false negative rate in unseen training examples. Thus, one can experiment the effect of training for a different number of iterations, which influences the size of the learned model.

Abbreviation	Description
AML	Algebraic Machine Learning
AI	Artificial Intelligence
DAW	Digital Audio Workstation
DL	Description Logic
DJ	Disc Jockey
EQ	Equalization (in audio)
FX	Effects (in audio processing)
GUI	Graphical User Interface
HCI	Human-Computer Interaction
ML	Machine Learning
MIDI	Musical Instrument Digital Interface
TUI	Tangible User Interface
BPM	Beats Per Minute
3D	Three-Dimensional
TS	TypeScript
ML	Multi-layer Perceptron
RQ	Research Question

Table D.1: Glossary of Short Forms